

Enabling Smart Cities with Renesas Smart Metering Solutions

Renesas is a full metering building block solution provider, from dedicated system-on-chip devices to reference designs with more than 30 years of metering experience.

Our latest RL78 core low power series 16-Bit microcontroller enable LED control, power supply control, and communication functions on a single chip and realize high processing performance while delivering the lowest power consumption. Its built-in RTC with hardware compensation and LCD driver not only help simplify circuit design, but also reduce production cost.

Renesas and partners enable fast time to market with off the shelf examples and solutions with multiple channels of support. Take advantage of our wide ranging collaborations and partnerships to jump start your next metering design.



RX600, RX200

Smart electricity meters and data concentrators

- High performance, high efficiency 32-bit MCU
- Ultra low power
- Powerful peripheral modules
- Integrated DSP functions
- Up to 2MB Flash with memory protection unit and security features
- Full Calendar RTC with Vbat

Single and 3-phase meters

- RX21A with built-in 24-bit Delta-sigma ADC
- Dedicated peripheral functions with special features, such as RTC
- Hardware encryption
- Temperature sensor

Data Collector & Housekeeping

- High performance RX core with up to 960 DMIPS
- Up to 2MB Flash and 256KB RAM
- Rich peripheral integration (Ethernet, Direct TFT drive)

RL78, RX100

Communications

- Advance single chip RF solutions
- Flexible multi chip solutions
- Highly robust PLC solutions:
 - support G3 and PRIME Standards with flexible software stack on same hardware
 - Security encrypt and decrypt functions based on AES
 - Integrated Analog Front End circuits, Zero cross circuit detection
 - Powerful DSP to handle PHY functions, with IEE 802.15.4 compliance MAC controller
- ZigBee™ + Smart Energy Profile (SEP1.1)
 - Extensive RAM/ROM/Flash Memory and peripheral combinations
 - Unmatched EM/EMS characteristics
 - ZigBee certifies IEEE802.15.4 Software Stack
- Ethernet

Gas, water and heat meters

- World leading low power MCU
- Low voltage
- Direct segment LCD drive
- Rich set of peripherals

<http://in.renesas.com/applications>

[Learn more!](#)